Lecture 19

Multi-Agent Learning

COMP532 – Machine Learning and Bioinspired Optimisation

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Today's Questions

At the end of this lecture, you should be able to answer:

1. Why is multi-agent learning fundamentally harder than single-agent RL?
2. What changes when multiple learners interact in the same environment?
3. What are the main mathematical tools we will need in this part of the module?

Why is Multi-Agent Learning Fundamentally Different?

Single-agent RL assumes a stationary environment.

Multi-agent learning breaks this assumption:

- Other agents are learning and adapting
- The environment becomes non-stationary
- Rewards depend on strategic interaction

This creates three core challenges:

1. **Prediction:** What will the other agents do?
2. **Coordination:** How can agents achieve good joint behaviour?
3. **Stability:** Will learning converge, oscillate, or fail?

Key Question

How should an agent learn when the environment itself is learning back?

From Single-Agent RL to Multi-Agent Learning

Single-agent RL

- One learner
- Environment dynamics are fixed
- Reward is defined by the task
- Optimality is usually individual

Multi-agent learning

- Multiple learners
- Dynamics depend on others' policies
- Reward depends on interaction
- Outcomes can involve
  - cooperation
  - competition
  - mixed motives

Takeaway

Multi-agent learning is not just RL with more agents. It requires reasoning about **interaction**, **adaptation**, and **equilibrium**.

A Simple Example: Why Other Agents Matter

Consider two agents choosing actions simultaneously.

	A	B
A	(3,3)	(0,1)
B	(1,0)	(2,2)

The return of each agent depends not only on its own action, but also on the other agent's action.

Key observations

- Best action is *context-dependent*
- Learning target changes if the opponent changes
- Exploration by one agent affects the other

Implication

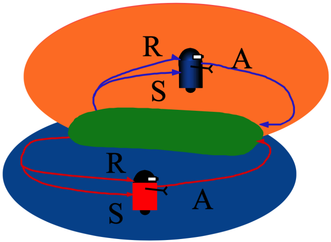
In multi-agent settings, agents are part of each other's environment.

What is Multi-Agent Learning?

Multi-Agent Learning (MAL) studies how multiple agents learn through interaction in a shared environment.

A typical MAL problem includes:

- a set of agents $\{1, \ldots, n\}$
- states $s \in \mathcal{S}$
- actions $a_i \in \mathcal{A}_i$ for each agent i
- rewards r_i for each agent
- transition dynamics possibly influenced by the *joint action*



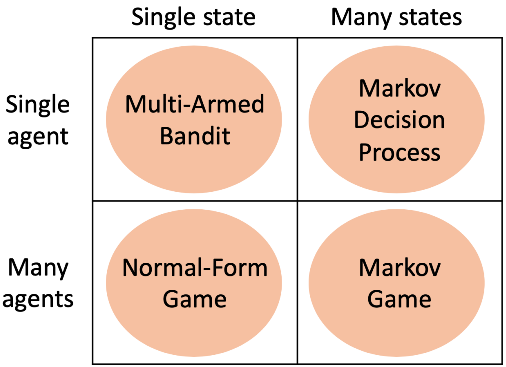
Central Theme

Each agent learns *while others are also learning*.

What Makes MAL Difficult?

1. **Non-stationarity**
   From one agent's perspective, the environment changes as others update their policies.
2. **Credit assignment**
   In cooperative tasks, how should reward be attributed to individual agents?
3. **Scalability**
   Joint action spaces often grow exponentially with the number of agents.
4. **Partial observability**
   Agents may not observe the full state or the internal state of other agents.
5. **Equilibrium and stability**
   Even if a desirable outcome exists, learning dynamics may not converge to it.

A Taxonomy of Learning Settings



- **Single agent, many states:** standard RL / MDPs
- **Multi-agent, single state:** matrix games / repeated games
- **Multi-agent, many states:** stochastic / Markov games

Three Major Interaction Regimes

Cooperative

- Shared objective
- Team reward
- Need coordination

Competitive

- Conflicting objectives
- Adversarial reasoning
- Robustness matters

Mixed-Motive

- Partial alignment
- Partial conflict
- Strategic complexity

Why this matters

The right learning objective and solution concept depend on the interaction structure.

Representative Application Domains

- **Robot teams:** coordination, task allocation, communication
- **Autonomous driving:** negotiation, collision avoidance, shared road use
- **Trading and markets:** strategic adaptation, competition, equilibrium
- **Network systems:** routing, resource allocation, congestion control
- **Games and simulation:** emergent strategies, cooperation, adversarial learning

Common feature

The behaviour of one learner changes the learning problem faced by others.

What Should an Agent Optimise?

In single-agent RL, we usually maximise expected return:

$$J(\pi) = \mathbb{E}\left[\sum_{t=0}^{\infty} \gamma^t r_t\right].$$

In multi-agent settings, this becomes more subtle:

- Should an agent maximise its own return?
- Should agents optimise a team objective?
- Should we seek a stable equilibrium?
- Should we optimise for robustness against adversaries?

Key Point

In MAL, the notion of “optimal” behaviour depends on how agents interact.

Why Game Theory Enters the Picture

To reason about strategic interaction, we need tools beyond standard RL.

Game theory provides:

- formal models of interaction
- concepts such as best response and Nash equilibrium
- a language for coordination and competition

But game theory alone is not enough:

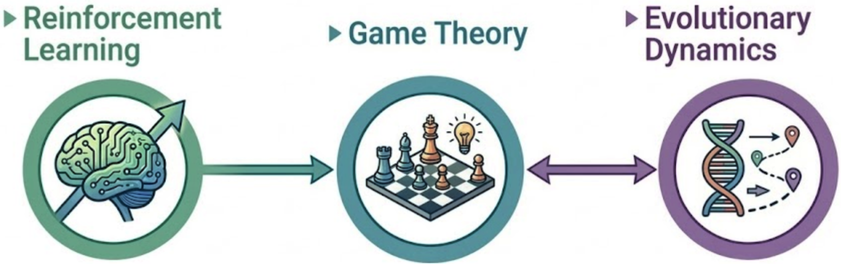
- it often studies *static* solution concepts
- it does not always explain *how agents learn*

This module will connect
RL + game theory + learning dynamics

Tools We Will Use in This Part of the Module

1. **Classical game theory**
   - normal-form games
   - best response
   - Nash equilibrium
2. **Stochastic / Markov games**
   - multi-agent extension of MDPs
   - sequential interaction
3. **Multi-agent RL algorithms**
   - independent learners
   - joint-action methods
   - equilibrium-based learning
4. **Evolutionary game theory**
   - population dynamics
   - replicator dynamics
   - stability analysis

A First High-Level Map of the Topic



- RL gives us learning algorithms
- game theory gives us strategic solution concepts
- evolutionary dynamics helps explain adaptation over time

What This Lecture Has Established

1. Multi-agent learning is difficult because agents learn in a **non-stationary strategic environment**.
2. The right notion of good behaviour depends on whether the setting is **cooperative, competitive, or mixed**.
3. To analyse MAL, we need a combination of
   - reinforcement learning,
   - game theory,
   - and learning dynamics.

Mental Model

Multi-agent learning = learning under **mutual adaptation**.

Looking Ahead

Next lectures:

- **Lecture 20:** Normal-form games, best response, Nash equilibrium
- **Lecture 21:** From static game analysis to learning behaviour
- **Lecture 22:** Multi-agent RL algorithms
- **Lecture 23–25:** Evolutionary game theory and links to RL

Bridge to the next lecture

Now that we understand why strategic interaction matters, we need a formal language to model it. That language starts with **game theory**.

Key Takeaways

- Multi-agent learning breaks the stationarity assumption of standard RL.
- Agents must reason about **others**, not just the environment.
- Strategic interaction requires tools from **game theory**.
- Learning over time motivates **dynamical** viewpoints as well.

Core message:

The environment is no longer passive — it learns back.